

Give every measure a contract

A decision you can explain

Connect the four branches below

1. Decompose the measure

A construct such as satisfaction becomes an observed variable through a stated question or measurement process. Coding does not automatically establish equal intervals.

2. Choose the right unit

Keep one observation at the intended grain. Define eligibility and dates before aggregating. A line-item join can duplicate an order-level outcome.

3. Show the denominator

The rate among measured cases answers a different question from the rate among all eligible cases. Unknown outcomes cannot simply be counted as successes.

4. Tell a bounded data story

Use bars for category comparisons, histograms for numeric distributions, and scatterplots for two numeric variables. Labels, units, source, and scale support honest comparison.

Decompose the measure

Construct > operational variable > values

The connection

A construct such as satisfaction becomes an observed variable through a stated question or measurement process. Coding does not automatically establish equal intervals.

A business example

A single satisfaction item is ordered categorical. Dollar revenue has a meaningful zero and supports ratios.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Choose the right unit

Grain > key > population > period

The connection

Keep one observation at the intended grain. Define eligibility and dates before aggregating. A line-item join can duplicate an order-level outcome.

A business example

An order with three products is still one order for an order-level late-delivery rate.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Show the denominator

Known outcomes > missing outcomes > bounds

The connection

The rate among measured cases answers a different question from the rate among all eligible cases. Unknown outcomes cannot simply be counted as successes.

A business example

Show the observed rate and plausible best/worst bounds when outcomes are missing.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Tell a bounded data story

Question > display > interpretation > action

The connection

Use bars for category comparisons, histograms for numeric distributions, and scatterplots for two numeric variables. Labels, units, source, and scale support honest comparison.

A business example

Start count bars at zero. Joining unordered categories with a line implies an unsupported ordering.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.